University of Chittagong

Dept. of Computer Science & Engineering

DBMS Assignment

Tutorial-02

Course Title: Database Management Systems
Course Code: CSE-414

Submitted to

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Task:01

1.Document and describe your database in the way you would do it (or have done it) in a project report.

Answer:

(I)Introduction: This database is designed to store and manage medical information related to drugs,disease and their relationships.The system is designed to support queries that analyze drug–disease relationships, identify common side effects, and explore clinical trial data efficiently so that users can retrieve meaningful insights about drugs and their applications.

(II)Scoop of the Database: The database includes the following types of information:

1.Drugs and their categories.

2.Diseases and their categories.

3.Side effects and Interactions of Drugs.

4.Commercial Drug Products and Companies.

5.Clinical trial information and their researcher and location.

(III)Assumption: To simplify the design, the following assumptions are made:

1.A drug can treat multiple diseases.

2.A disease can be treated by multiple drugs.

3.A drug can have multiple side effects.

4.A side effect can be caused by multiple drugs.

5.Drug interactions occur between two or more drugs.

6.A drug may have multiple commercial products.

7.A drug may be involved in multiple clinical trials.

(IV)Entities & Relationship

Entities:

1.Drug

2.Disease

3.Side-effect

4.Clinical Trial

5.Researcher

6.product

7.Company

Relationship:

1.Drugs and Diseases(M:N)

2.Drugs and Side Effects(1:N)

3.Drugs and Clinical Trials(M:N)

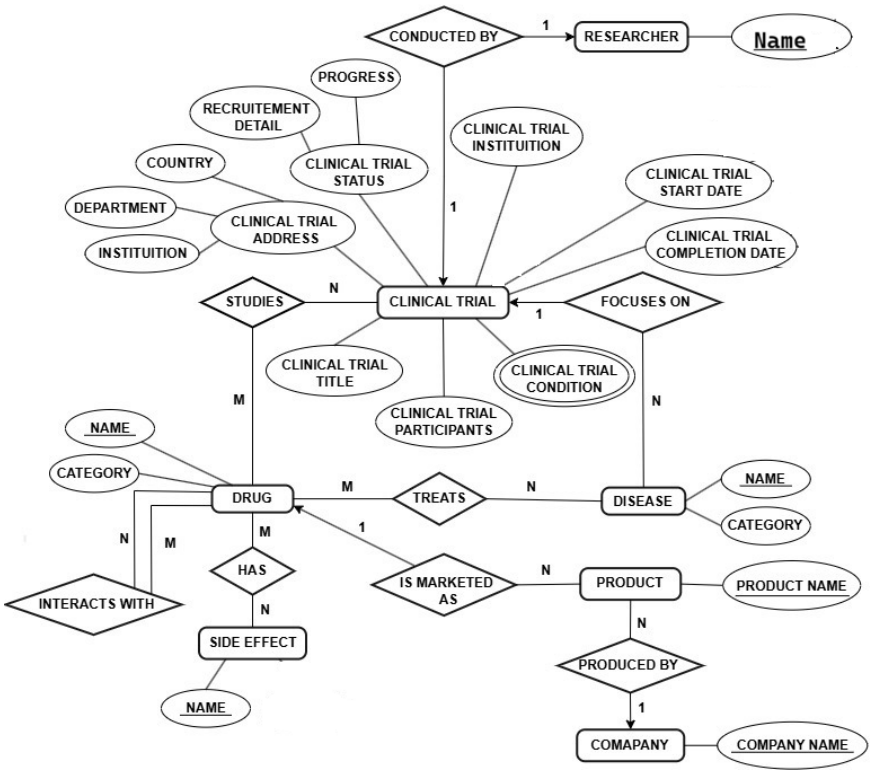
4.Drugs can interact with other Drugs (self-relationship)

5.Drugs and Products(1:N)

6.Products and company(N:1)

7.Clinical trials and Researcher(1:1)

2. Provide a list of tables with attributes that you would use to store your data.


CONDUCTED BY
1
RESEARCHER
Name
PROGRESS
RECRUITEMENT DETAIL
CLINICAL TRIAL INSTITUITION
COUNTRY
CLINICAL TRIAL STATUS
CLINICAL TRIAL START DATE
DEPARTMENT
CLINICAL TRIAL ADDRESS
1
CLINICAL TRIAL COMPLETION DATE
INSTITUITION
STUDIES
N
CLINICAL TRIAL
FOCUSES ON
1
CLINICAL TRIAL TITLE
CLINICAL TRIAL CONDITION
M
N
CLINICAL TRIAL PARTICIPANTS
NAME
CATEGORY
DRUG
M
TREATS
N
DISEASE
NAME
CATEGORY
N
M
M
1
HAS
IS MARKETED AS
N
PRODUCT
PRODUCT NAME
INTERACTS WITH
N
SIDE EFFECT
N
PRODUCED BY
NAME
1
COMAPANY
COMPANY NAME

3. In case you already have a good command of SQL, add the SQL commands that create your database.

Answer:

DATABASE

```sql
DROP DATABASE IF EXISTS PHARMA_DB;
CREATE DATABASE IF NOT EXISTS PHARMA_DB;
USE PHARMA_DB;

-- ==========================
-- MAIN ENTITY TABLES
-- ==========================

CREATE TABLE IF NOT EXISTS DRUG (
    DRUG_NAME VARCHAR(255) PRIMARY KEY,
    DRUG_CATEGORY VARCHAR(255)
);

CREATE TABLE IF NOT EXISTS DISEASE (
    DISEASE_NAME VARCHAR(255) PRIMARY KEY,
    DISEASE_CATEGORY VARCHAR(255)
);

CREATE TABLE IF NOT EXISTS SIDE_EFFECT (
    SIDE_EFFECT_NAME VARCHAR(255) PRIMARY KEY
);

CREATE TABLE IF NOT EXISTS RESEARCHER (
    MAIN_RESEARCHER VARCHAR(255) PRIMARY KEY
);

CREATE TABLE IF NOT EXISTS COMPANY (
    COMPANY_NAME VARCHAR(255) PRIMARY KEY
);

CREATE TABLE IF NOT EXISTS PRODUCT (
    PRODUCT_NAME VARCHAR(255) PRIMARY KEY
);

CREATE TABLE IF NOT EXISTS CLINICAL_TRIAL (
    CLINICAL_TRIAL_TITLE VARCHAR(500) PRIMARY KEY,
    CLINICAL_TRIAL_STATUS VARCHAR(100),
    CLINICAL_TRIAL_COUNTRY VARCHAR(255),
    CLINICAL_TRIAL_DEPARTMENT VARCHAR(255),
    CLINICAL_TRIAL_INSTITUTION VARCHAR(255),
    CLINICAL_TRIAL_START_DATE DATE,
    CLINICAL_TRIAL_COMPLETION_DATE DATE,
    CLINICAL_TRIAL_CONDITION VARCHAR(255),
    CLINICAL_TRIAL_PARTICIPANTS INT
);
```

```sql
-- ========================
-- RELATIONSHIP TABLES
-- ========================
CREATE TABLE IF NOT EXISTS STUDIES (
    CLINICAL_TRIAL_TITLE VARCHAR(500),
    DRUG_NAME VARCHAR(255),
    PRIMARY KEY (CLINICAL_TRIAL_TITLE, DRUG_NAME),
    CONSTRAINT FK_STUDIES_TRIAL FOREIGN KEY (CLINICAL_TRIAL_TITLE)
        REFERENCES CLINICAL_TRIAL(CLINICAL_TRIAL_TITLE)
        ON DELETE CASCADE
        ON UPDATE CASCADE,
    CONSTRAINT FK_STUDIES_DRUG FOREIGN KEY (DRUG_NAME)
        REFERENCES DRUG(DRUG_NAME)
        ON DELETE CASCADE
        ON UPDATE CASCADE
);
-- TREATS : DRUG (M) <-> (N) DISEASE
CREATE TABLE IF NOT EXISTS TREATS (
    DRUG_NAME VARCHAR(255),
    DISEASE_NAME VARCHAR(255),
    PRIMARY KEY (DRUG_NAME, DISEASE_NAME),
    CONSTRAINT FK_TREATS_DRUG FOREIGN KEY (DRUG_NAME)
        REFERENCES DRUG(DRUG_NAME)
        ON DELETE CASCADE
        ON UPDATE CASCADE,
    CONSTRAINT FK_TREATS_DISEASE FOREIGN KEY (DISEASE_NAME)
        REFERENCES DISEASE(DISEASE_NAME)
        ON DELETE CASCADE
        ON UPDATE CASCADE
);
-- HAS : DRUG (M) <-> (N) SIDE_EFFECT
CREATE TABLE IF NOT EXISTS HAS_SIDE_EFFECT (
    DRUG_NAME VARCHAR(255),
    SIDE_EFFECT_NAME VARCHAR(255),
    PRIMARY KEY (DRUG_NAME, SIDE_EFFECT_NAME),
    CONSTRAINT FK_HAS_DRUG FOREIGN KEY (DRUG_NAME)
        REFERENCES DRUG(DRUG_NAME)
        ON DELETE CASCADE
        ON UPDATE CASCADE,
    CONSTRAINT FK_HAS_SIDE_EFFECT FOREIGN KEY (SIDE_EFFECT_NAME)
        REFERENCES SIDE_EFFECT(SIDE_EFFECT_NAME)
        ON DELETE CASCADE
        ON UPDATE CASCADE
);
-- INTERACTS_WITH : recursive M:N on DRUG
CREATE TABLE IF NOT EXISTS INTERACTS_WITH (
    DRUG_NAME VARCHAR(255),
    INTERACTS_WITH_DRUG_NAME VARCHAR(255),
    PRIMARY KEY (DRUG_NAME, INTERACTS_WITH_DRUG_NAME),
    CONSTRAINT FK_INTERACTS_DRUG1 FOREIGN KEY (DRUG_NAME)
        REFERENCES DRUG(DRUG_NAME)
```

```
        ON DELETE CASCADE
        ON UPDATE CASCADE,
    CONSTRAINT FK_INTERACTS_DRUG2 FOREIGN KEY (INTERACTS_WITH_DRUG_NAME)
        REFERENCES DRUG(DRUG_NAME)
        ON DELETE CASCADE
        ON UPDATE CASCADE
);
-- CONDUCTED_BY : CLINICAL_TRIAL (1) <-> (1) RESEARCHER
CREATE TABLE IF NOT EXISTS CONDUCTED_BY (
    CLINICAL_TRIAL_TITLE VARCHAR(500) PRIMARY KEY,
    MAIN_RESEARCHER VARCHAR(255) UNIQUE,
    CONSTRAINT FK_CB_TRIAL FOREIGN KEY (CLINICAL_TRIAL_TITLE)
        REFERENCES CLINICAL_TRIAL(CLINICAL_TRIAL_TITLE)
        ON DELETE CASCADE
        ON UPDATE CASCADE,
    CONSTRAINT FK_CB_RESEARCHER FOREIGN KEY (MAIN_RESEARCHER)
        REFERENCES RESEARCHER(MAIN_RESEARCHER)
        ON DELETE CASCADE
        ON UPDATE CASCADE
);
-- FOCUSES_ON : CLINICAL_TRIAL (1) -> (N) DISEASE
-- One clinical trial focuses on one disease entry here
-- A disease can appear in many trials
CREATE TABLE IF NOT EXISTS FOCUSES_ON (
    CLINICAL_TRIAL_TITLE VARCHAR(500) PRIMARY KEY,
    DISEASE_NAME VARCHAR(255),
    CONSTRAINT FK_FO_TRIAL FOREIGN KEY (CLINICAL_TRIAL_TITLE)
        REFERENCES CLINICAL_TRIAL(CLINICAL_TRIAL_TITLE)
        ON DELETE CASCADE
        ON UPDATE CASCADE,
    CONSTRAINT FK_FO_DISEASE FOREIGN KEY (DISEASE_NAME)
        REFERENCES DISEASE(DISEASE_NAME)
        ON DELETE CASCADE
        ON UPDATE CASCADE
);
-- IS_MARKETED_AS : DRUG (1) -> (N) PRODUCT
-- One drug can have many products
-- Each product belongs to one drug
CREATE TABLE IF NOT EXISTS IS_MARKETED_AS (
    PRODUCT_NAME VARCHAR(255) PRIMARY KEY,
    DRUG_NAME VARCHAR(255),
    CONSTRAINT FK_IMA_PRODUCT FOREIGN KEY (PRODUCT_NAME)
        REFERENCES PRODUCT(PRODUCT_NAME)
        ON DELETE CASCADE
        ON UPDATE CASCADE,
    CONSTRAINT FK_IMA_DRUG FOREIGN KEY (DRUG_NAME)
        REFERENCES DRUG(DRUG_NAME)
        ON DELETE CASCADE
        ON UPDATE CASCADE
);
-- PRODUCED_BY : PRODUCT (N) -> (1) COMPANY
```

```sql
CREATE TABLE IF NOT EXISTS PRODUCED_BY (
    PRODUCT_NAME VARCHAR(255) PRIMARY KEY,
    COMPANY_NAME VARCHAR(255),
    CONSTRAINT FK_PB_PRODUCT FOREIGN KEY (PRODUCT_NAME)
        REFERENCES PRODUCT(PRODUCT_NAME)
        ON DELETE CASCADE
        ON UPDATE CASCADE,
    CONSTRAINT FK_PB_COMPANY FOREIGN KEY (COMPANY_NAME)
        REFERENCES COMPANY(COMPANY_NAME)
        ON DELETE CASCADE
        ON UPDATE CASCADE
);
```

Python SCRIPT

```
import pandas as pd
import os
import mysql.connector
from mysql.connector import Error
from collections import defaultdict

# ========== CONFIGURATION ==========
# EXCEL_FILE = "sample.xlsx"
BASE_DIR = os.path.dirname(os.path.abspath(__file__))
EXCEL_FILE = os.path.join(BASE_DIR, "sample.xlsx")
DB_CONFIG = {
    "host": "localhost",
    "user": "sheikh", # change as needed
    "password": "miniso123", # set your MySQL root password
    "database": "PHARMA_DB_TASK1"
}

# ========== HELPER FUNCTIONS ==========
def clean_str(val):
    """Convert to string, strip, return None if empty or NaN."""
    if pd.isna(val):
        return None
    s = str(val).strip()
    return s if s else None

def get_side_effects(row):
    """Extract non-empty side effects from columns B-F (indices 1-5)."""
    effects = []
    for i in range(1, 6): # columns B to F
        eff = clean_str(row.iloc[i])
        if eff:
            effects.append(eff)
    return effects

def get_interacts(row):
    """Extract non-empty interacting drugs from columns G-I (indices 6-8)."""
    interacts = []
    for i in range(6, 9):
        drug = clean_str(row.iloc[i])
        if drug:
            interacts.append(drug)
    return interacts

# ========== MAIN IMPORT ==========
def main():
    # 1. Read Excel
    df = pd.read_excel(EXCEL_FILE, sheet_name="Sheet1", header=0)
    # Rename columns for easier access (optional)
    # Expected columns order:
    # 0:A drug_name, 1:B side_effect1, 2:C side_effect2, 3:D side_effect3, 4:E
```

```
    side_effect4, 5:F side_effect5,
# 6:G interacts1, 7:H interacts2, 8:I interacts3, 9:J disease_name, 10:K
    disease_category,
# 11:L drug_category, 12:M product_name, 13:N company_name, 14:O
    clinical_trial_title,
# 15:P start_date, 16:Q completion_date, 17:R participants, 18:S status,
# 19:T condition1, 20:U condition2, 21:V address/country, 22:W institution,
# 23:X department, 24:Y main_researcher, 25:Z condition3
# We'll use iloc positions.

# Data containers
drugs = {} # drug_name -> set of categories
diseases = {} # disease_name -> disease_category
side_effects = set()
researchers = set()
companies = set()
products = {} # product_name -> (drug_name, company_name)
clinical_trials = {} # trial_title -> dict of attributes
interacts_set = set() # (drug_name, interacts_with)
treats_set = set() # (drug_name, disease_name)
has_side_effect_set = set() # (drug_name, side_effect)
studies_set = set() # (trial_title, drug_name)
conducted_by_dict = {} # trial_title -> researcher
focuses_on_dict = {} # trial_title -> disease_name (first condition)
is_marketed_as_dict = {} # product_name -> drug_name
produced_by_dict = {} # product_name -> company_name

# 2. Iterate rows
for idx, row in df.iterrows():
    drug_name = clean_str(row.iloc[0])
    if not drug_name:
        # Skip rows without a drug name (they are incomplete/duplicate
            entries)
        continue

    # --- Drug & Category ---
    drug_cat = clean_str(row.iloc[11])
    if drug_cat:
        if drug_name not in drugs:
            drugs[drug_name] = set()
        drugs[drug_name].add(drug_cat)

    # --- Disease & Category ---
    disease_name = clean_str(row.iloc[9])
    disease_cat = clean_str(row.iloc[10])
    if disease_name:
        diseases[disease_name] = disease_cat
        # Treats relationship
        treats_set.add((drug_name, disease_name))

    # --- Side Effects ---
```

```
        for effect in get_side_effects(row):
            side_effects.add(effect)
            has_side_effect_set.add((drug_name, effect))

        # --- Interacts With ---
        for interact_drug in get_interacts(row):
            # Ensure the interacting drug exists in drugs set (will be added
                later)
            # We will add it to drugs with unknown category later.
            if interact_drug not in drugs:
                drugs[interact_drug] = set() # placeholder, category unknown
            interacts_set.add((drug_name, interact_drug))

        # --- Product & Company ---
        product_name = clean_str(row.iloc[12])
        company_name = clean_str(row.iloc[13])
        if product_name and company_name:
            products[product_name] = (drug_name, company_name)
            is_marketed_as_dict[product_name] = drug_name
            produced_by_dict[product_name] = company_name
            companies.add(company_name)

        # --- Clinical Trial ---
        trial_title = clean_str(row.iloc[14])
        if trial_title:
            # Check if we already processed this trial
            if trial_title not in clinical_trials:
                # Extract trial attributes
                start = row.iloc[15]
                completion = row.iloc[16]
                participants = row.iloc[17]
                status = clean_str(row.iloc[18])
                country = clean_str(row.iloc[21]) # column V
                institution = clean_str(row.iloc[22]) # column W
                department = clean_str(row.iloc[23]) # column X
                # Convert dates
                start_date = pd.to_datetime(start).date() if pd.notna(start) else
                    None
                end_date = pd.to_datetime(completion).date() if pd.notna(
                    completion) else None
                participants_int = int(participants) if pd.notna(participants)
                    else None
                # First condition (column T) for FOCUSES_ON
                condition1 = clean_str(row.iloc[19])
                clinical_trials[trial_title] = {
                    "status": status,
                    "country": country,
                    "department": department,
                    "institution": institution,
                    "start_date": start_date,
                    "completion_date": end_date,
```

```
                "condition": condition1, # for focus
                "participants": participants_int
            }
        else:
            # If trial already exists, maybe update condition? Use first
                encountered.
            pass

        # Studies relationship
        studies_set.add((trial_title, drug_name))

        # Conducted By (researcher)
        researcher = clean_str(row.iloc[24])
        if researcher:
            researchers.add(researcher)
            conducted_by_dict[trial_title] = researcher

        # Focuses On (use condition from column T, only if not already set)
        # Use the first condition we see; if already set, keep original.
        if trial_title not in focuses_on_dict:
            cond = clean_str(row.iloc[19])
            if cond:
                focuses_on_dict[trial_title] = cond

# 3. Insert into MySQL
try:
    conn = mysql.connector.connect(**DB_CONFIG)
    cursor = conn.cursor()
    # Disable foreign key checks for bulk insert order
    cursor.execute("SET FOREIGN_KEY_CHECKS = 0")

    # ---- Insert DRUGS (with concatenated categories) ----
    for drug_name, cat_set in drugs.items():
        cat_str = ", ".join(sorted(cat_set)) if cat_set else None
        cursor.execute(
            "INSERT IGNORE INTO DRUG (DRUG_NAME, DRUG_CATEGORY) VALUES (%s, %
                s)",
            (drug_name, cat_str)
        )

    # ---- Insert DISEASE ----
    for disease_name, disease_cat in diseases.items():
        cursor.execute(
            "INSERT IGNORE INTO DISEASE (DISEASE_NAME, DISEASE_CATEGORY)
                VALUES (%s, %s)",
            (disease_name, disease_cat)
        )

    # ---- Insert SIDE_EFFECT ----
    for effect in side_effects:
        cursor.execute(
```

```
        "INSERT IGNORE INTO SIDE_EFFECT (SIDE_EFFECT_NAME) VALUES (%s)",
        (effect,)
    )

# ---- Insert RESEARCHER ----
for researcher in researchers:
    cursor.execute(
        "INSERT IGNORE INTO RESEARCHER (MAIN_RESEARCHER) VALUES (%s)",
        (researcher,)
    )

# ---- Insert COMPANY ----
for company in companies:
    cursor.execute(
        "INSERT IGNORE INTO COMPANY (COMPANY_NAME) VALUES (%s)",
        (company,)
    )

# ---- Insert PRODUCT ----
for product_name in products.keys():
    cursor.execute(
        "INSERT IGNORE INTO PRODUCT (PRODUCT_NAME) VALUES (%s)",
        (product_name,)
    )

# ---- Insert CLINICAL_TRIAL ----
for title, attrs in clinical_trials.items():
    cursor.execute("""
        INSERT IGNORE INTO CLINICAL_TRIAL (
            CLINICAL_TRIAL_TITLE, CLINICAL_TRIAL_STATUS,
                CLINICAL_TRIAL_COUNTRY,
            CLINICAL_TRIAL_DEPARTMENT, CLINICAL_TRIAL_INSTITUTION,
            CLINICAL_TRIAL_START_DATE, CLINICAL_TRIAL_COMPLETION_DATE,
            CLINICAL_TRIAL_CONDITION, CLINICAL_TRIAL_PARTICIPANTS
        ) VALUES (%s, %s, %s, %s, %s, %s, %s, %s, %s)
    """, (
        title, attrs["status"], attrs["country"],
        attrs["department"], attrs["institution"],
        attrs["start_date"], attrs["completion_date"],
        attrs["condition"], attrs["participants"]
    ))

# ---- Relationship tables ----
# TREATS
for drug_name, disease_name in treats_set:
    cursor.execute(
        "INSERT IGNORE INTO TREATS (DRUG_NAME, DISEASE_NAME) VALUES (%s,
            %s)",
        (drug_name, disease_name)
    )
```

```
# HAS_SIDE_EFFECT
for drug_name, effect in has_side_effect_set:
    cursor.execute(
        "INSERT IGNORE INTO HAS_SIDE_EFFECT (DRUG_NAME, SIDE_EFFECT_NAME)
            VALUES (%s, %s)",
        (drug_name, effect)
    )

# INTERACTS_WITH
for drug_name, interact_drug in interacts_set:
    cursor.execute(
        "INSERT IGNORE INTO INTERACTS_WITH (DRUG_NAME,
            INTERACTS_WITH_DRUG_NAME) VALUES (%s, %s)",
        (drug_name, interact_drug)
    )

# STUDIES
for trial_title, drug_name in studies_set:
    cursor.execute(
        "INSERT IGNORE INTO STUDIES (CLINICAL_TRIAL_TITLE, DRUG_NAME)
            VALUES (%s, %s)",
        (trial_title, drug_name)
    )

# CONDUCTED_BY
for trial_title, researcher in conducted_by_dict.items():
    cursor.execute(
        "INSERT IGNORE INTO CONDUCTED_BY (CLINICAL_TRIAL_TITLE,
            MAIN_RESEARCHER) VALUES (%s, %s)",
        (trial_title, researcher)
    )

# FOCUSES_ON
for trial_title, disease_name in focuses_on_dict.items():
    cursor.execute(
        "INSERT IGNORE INTO FOCUSES_ON (CLINICAL_TRIAL_TITLE,
            DISEASE_NAME) VALUES (%s, %s)",
        (trial_title, disease_name)
    )

# IS_MARKETED_AS
for product_name, drug_name in is_marketed_as_dict.items():
    cursor.execute(
        "INSERT IGNORE INTO IS_MARKETED_AS (PRODUCT_NAME, DRUG_NAME)
            VALUES (%s, %s)",
        (product_name, drug_name)
    )

# PRODUCED_BY
for product_name, company_name in produced_by_dict.items():
    cursor.execute(
```

```
                "INSERT IGNORE INTO PRODUCED_BY (PRODUCT_NAME, COMPANY_NAME)
                    VALUES (%s, %s)",
                (product_name, company_name)
            )

        # Re-enable foreign key checks
        cursor.execute("SET FOREIGN_KEY_CHECKS = 1")
        conn.commit()
        print("Data imported successfully!")

    except Error as e:
        print(f"Error: {e}")
        conn.rollback()
    finally:
        if conn.is_connected():
            cursor.close()
            conn.close()

if __name__ == "__main__":
    main()
```

Task:02

1. Identify relevant data (consider the sample queries included in this document).

Answer

To design the database, the relevant data is identified based on the requirements and sample queries. The goal is to determine what information must be stored so that all required queries can be answered efficiently.

→ Identified Data Entities and Attributes

1.Drug

drug_name(PK)

drug_category

interacts_with

*Needed for: almost all queries (a, b, c, d, e, f, g, h, i)

2.Side Effect

side_effect_name(PK)

*Needed for:

(a) nausea

(c) cough and headache

(g) top side effects

3.Disease

disease_name(PK)

disease_category

Needed for:

(d) endocrine diseases

(e) immunological vs hematological

(f) hydrocortisone vs etanercept

(j, k) respiratory & cardiovascular

4.Product

product_name(PK)

company_name

5.Clinical Trial

title(PK)

status

start_date

-Needed for query (i)

completion_date

-Needed for query (i)

participants

-Needed for query (h)

condition

address(Composite attribute: country, department, institution)

-Needed for query (l)

6.Researcher

Name(PK)

Relationship	**Entities Involved**	**Cardinality**	**Queries Supported**
treats	Drug ↔ Disease	M:N	(d), (e), (f), (g)
has	Drug ↔ Side Effect	M:N	(a), (c), (g)
is marketed as	Drug → Product	1:N	commercial product lookup
interacts with	Drug ↔ Drug	M:N (self-ref)	(b)
studies	Clinical Trial ↔ Drug	M:N	(h), (i), (j), (k)
conducted by	Clinical Trial → Researcher	N:1	(j), (k)